



Métodos de diferenças finitas.

Derivadas de 1 orden nos espaço? Erro nas discretizações das Derivadas espaciais

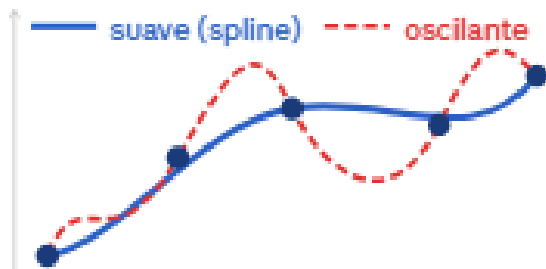
Faça a Expansão de Taylor para 1 ponto de grade



Restrições Adicionais da Função Interpoladora

Além de passar pelos pontos dados, podem-se impor restrições adicionais à função interpolante: **suavidade**, **monotonicidade** ou **convexidade**. Essas propriedades garantem que a solução seja fisicamente plausível.

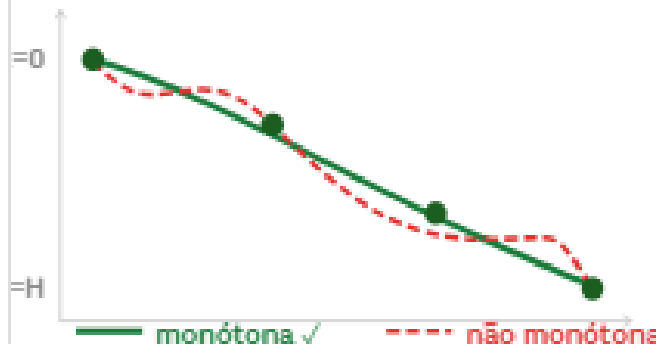
Suavidade



f é suave se suas derivadas são contínuas no intervalo.
Formalmente: $f \in C^k[a, b]$.
Garante ausência de descontinuidades artificiais.

Aplicação: campos de vento e temperatura em modelos atmosféricos devem ser contínuos — evitar gradientes espúrios que disparam instabilidades numéricas.

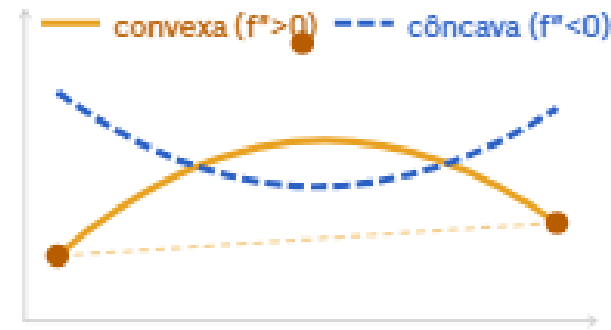
Monotonicidade



f é monótona se $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$ (crescente) ou $f(x_1) \geq f(x_2)$ (decrescente) em todo o intervalo.

Aplicação: perfil de densidade oceânica deve ser monotônico com a profundidade (coluna estável). Inversões espúrias geram instabilidade convectiva artificial.

Convexidade



f é convexa se $f''(x) \geq 0$ (curva abaixo da secante); côncava se $f''(x) \leq 0$.
Restringe o formato global da interpolante.

Aplicação: perfil de umidade relativa na troposfera livre tende a ser côncavo. Preservar convexidade/concavidade evita supersaturação espúria em níveis intermediários.

⚠ **Atenção:** impor restrições adicionais aumenta a complexidade do método (ex: splines monotônicas de Fritsch-Carlson, interpolação PCHIP). Em modelos operacionais, o balanço entre fidelidade aos dados e consistência física é fundamental.



Métodos de diferenças finitas.

Série de fourier

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

Faça a Expansão de Taylor para 1 ponto de grade

Serie de taylor

$$f(x) = f(a)(x-a)^0 + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^n(a)(x-a)^n}{n!}$$

l,m						n,m
		$i-1,j+1$	$i,j+1$	$i+1,j+1$		
		$i-1,j$	i,j	$i+1,j$		
		$i-1,j-1$	$i,j-1$	$i+1,j-1$		
l,l						n,l

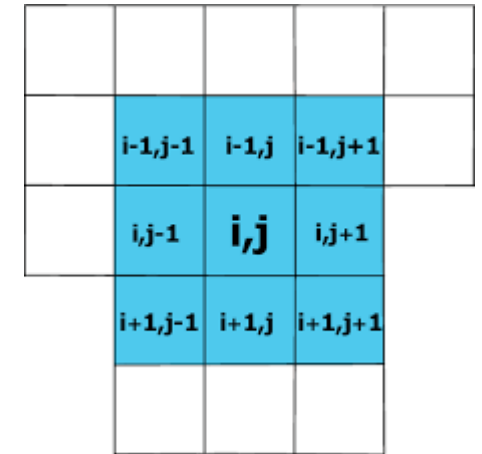


Métodos de diferenças finitas.

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

A partir da eq 1B e 2B, obtém-se Discretização de 1 ordem o erro



$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{u_{j+1}^n - u_j^n}{\Delta x} = \frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$



Métodos de diferenças finitas.

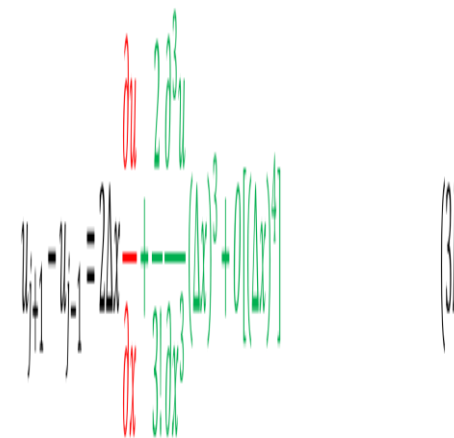
$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

Subtrai 1B-2B , obtem-se a Discretização de 2 ordem no erro

$$u_{j+1} - u_{j-1} = 2\Delta x \frac{\partial u}{\partial x} + \frac{2}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^3 + O[(\Delta x)^4] \quad (3B)$$

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{1}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O[(\Delta x)^4] \quad (4B)$$





Métodos de diferenças finitas.

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	

soma 1B+2B, , obtém-se a Discretização de 2 ordem na derivada e 2 ordem no erro

$$u_{j+1} + u_{j-1} = 2u_j + \frac{\partial^2 u}{\partial x^2} \Delta x^2 + \frac{2}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^4 + O[(\Delta x)^4] \quad (5B)$$

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} = + \frac{\partial^2 u}{\partial x^2} + \frac{2}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6B)$$



Métodos de diferenças finitas.

Faça a Expansão de Taylor para 2 ponto de grade

Serie de taylor

$$f(x) = f(a)(x-a)^0 + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^n(a)(x-a)^n}{n!}$$

l,m						n,m
		$i-1,j+1$	$i,j+1$	$i+1,j+1$		
		$i-1,j$	i,j	$i+1,j$		
		$i-1,j-1$	$i,j-1$	$i+1,j-1$		
l,l						n,l



Métodos de diferenças finitas.

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$

A partir da eq 1c e 2C, obtém-se Discretização de 1 ordem o erro

$$\rightarrow \frac{u_{j+2}^n - u_j^n}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{6\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{14\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\rightarrow \frac{(u_j^n - u_{j-2}^n)}{2\Delta x} = \frac{\partial u}{\partial x} - \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{6\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	



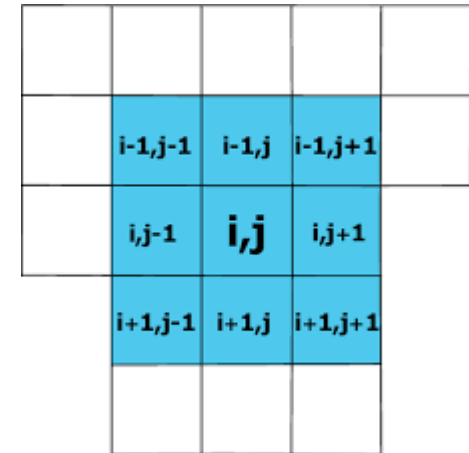
Métodos de diferenças finitas.

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$

Subtrai 1c-2C , obtém-se a Discretização de 2 ordem no erro

$$u_{j+2} - u_{j-2} = 4\Delta x \frac{\partial u}{\partial x} + \frac{16}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^3 + O[(\Delta x)^4] \quad (3C)$$



$$\frac{u_{j+2} - u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + \frac{4}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O[(\Delta x)^4] \quad (4C)$$



Métodos de diferenças finitas.

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$

soma 1C+2C, , obtém-se a Discretização de 2 ordem na derivada e 2 orden no erro

$$u_{j+2} + u_{j-2} = 2u_j + \frac{8\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{32}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^4 + O[(\Delta x)^4]$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{8\Delta x^2} = + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$



Métodos de diferenças finitas.

Combinação das a Expansão de Taylor para 1 e 2 ponto de grade

segunda ordem

$$f(x) = f(a)(x-a)^0 + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^n(a)(x-a)^n}{n!}$$

l, m						n, m
		$i-1, j+1$	$i, j+1$	$i+1, j+1$		
		$i-1, j$	i, j	$i+1, j$		
		$i-1, j-1$	$i, j-1$	$i+1, j-1$		
$l, 1$						$n, 1$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

Expansão de Taylor para 1 ponto de grade

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

Expansão de Taylor para 2 ponto de grade

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$

Rearranje os termos das equações

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_j^n - u_{j-2}^n)}{2\Delta x} = \frac{\partial u}{\partial x} - \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_j^n - u_{j-2}^n)}{2\Delta x} = \frac{\partial u}{\partial x} - \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

Multiplica a BB por 2

$$2 \left(\frac{u_j - u_{j-1}}{\Delta x} \right) = 2 \frac{\partial u}{\partial x} - 2 \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{2\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{2\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} + O[(\Delta x)^4]$$

soma a AD a eq. CC

$$2 \left(\frac{u_j - u_{j-1}}{\Delta x} \right) + \frac{u_j - u_{j-2}}{2\Delta x} = 3 \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \frac{6\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} \dots$$

$$\frac{1}{3} \left(\frac{2u_j - 2u_{j-1}}{\Delta x} \right) + \frac{1}{3} \frac{u_j - u_{j-2}}{2\Delta x} = \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

$$\frac{1}{3} \left(\frac{2 * 2u_j - 2 * 2u_{j-1} + u_j - u_{j-2}}{2\Delta x} \right) = \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

	i-1,j-1	i-1,j	i-1,j+1	
	i,j-1	i,j	i,j+1	
	i+1,j-1	i+1,j	i+1,j+1	



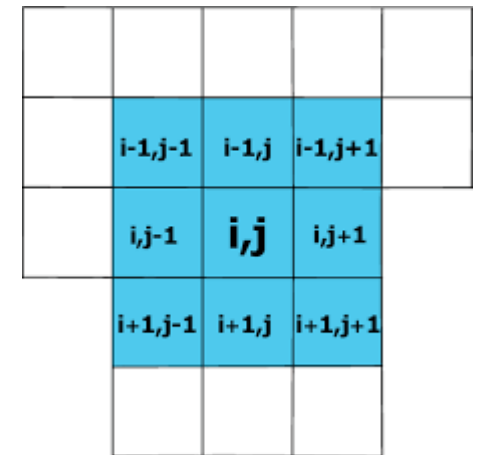
Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\frac{1}{3} \left(\frac{2u_j - 2u_{j-1}}{\Delta x} \right) + \frac{1}{3} \frac{u_j - u_{j-2}}{2\Delta x} = \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

$$\frac{1}{3} \left(\frac{2 * 2u_j - 2 * 2u_{j-1} + u_j - u_{j-2}}{2\Delta x} \right) = \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

$$\frac{1}{3} \left(\frac{5u_j - 4u_{j-1} - u_{j-2}}{2\Delta x} \right) = \frac{\partial u}{\partial x} - 2\Delta x \frac{\partial^2 u}{\partial x^2} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$





Métodos de diferenças finitas.

Combinação das a Expansão de Taylor para 1 e 2 ponto de grade

2 opção segunda ordem

$$f(x) = f(a)(x-a)^0 + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^n(a)(x-a)^n}{n!}$$

l,m						n,m
		$i-1,j+1$	$i,j+1$	$i+1,j+1$		
		$i-1,j$	i,j	$i+1,j$		
		$i-1,j-1$	$i,j-1$	$i+1,j-1$		
l,l						n,l



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

Expansão de Taylor para 1 ponto de grade

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

Expansão de Taylor para 2 ponto de grade

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\rightarrow \frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\rightarrow \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

Multiplica a BB por 2

$$2 \frac{u_{j-1} - u_j}{\Delta x} = -2 \frac{\partial u}{\partial x} + 2 \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{2\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{2\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} + O[(\Delta x)^4] \quad (AD)$$

subtrai a AD a eq. CC

$$2 \frac{u_{j-1} - u_j}{\Delta x} - \frac{u_{j+2} - u_j}{2\Delta x} = -3 \frac{\partial u}{\partial x} - \frac{6\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} \dots$$

$$\frac{2u_{j-1} - 2u_j}{3\Delta x} - \frac{u_{j+2} - u_j}{6\Delta x} = \frac{\partial u}{\partial x} - \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

$$\left(\frac{4u_{j-1} - 4u_j - u_{j+2} + u_j}{6\Delta x} \right) = \frac{\partial u}{\partial x} + \frac{\Delta x^2}{3} \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\left(\frac{4u_{j-1} - 4u_j - u_{j+2} + u_j}{6\Delta x} \right) = \frac{\partial u}{\partial x} + \frac{\Delta x^2}{3} \frac{\partial^3 u}{\partial x^3} \dots\dots (2D)$$

$$\left(\frac{4u_{j-1} - 4u_j - u_{j+2} + u_j}{6\Delta x} \right) = \frac{\partial u}{\partial x} + \frac{\Delta x^2}{3} \frac{\partial^3 u}{\partial x^3} \dots\dots (2D)$$

$$\left(\frac{4u_{j-1} - 3u_j - u_{j+2}}{6\Delta x} \right) = \frac{\partial u}{\partial x} + \frac{\Delta x^2}{3} \frac{\partial^3 u}{\partial x^3} \dots\dots (2D)$$



Métodos de diferenças finitas.



Combinação das a Expansão de Taylor para 1 e 2 ponto de grade

segunda ordem

Avançado

$$f(x) = f(a)(x-a)^0 + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^n(a)(x-a)^n}{n!}$$

l, m						n, m
		$i-1, j+1$	$i, j+1$	$i+1, j+1$		
		$i-1, j$	i, j	$i+1, j$		
		$i-1, j-1$	$i, j-1$	$i+1, j-1$		
l, l						n, l



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

Expansão de Taylor para 1 ponto de grade

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

Expansão de Taylor para 2 ponto de grade

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\frac{u_{j+1}^n - u_j^n}{\Delta x} = \frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

Multiplica por 4

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} = -4 \frac{\partial u}{\partial x} + \frac{4\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (DD)$$

Subtra DD da CC

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = -5 \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} - \Delta x \frac{\partial^2 u}{\partial x^2} = -5 \frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} - \Delta x \frac{\partial^2 u}{\partial x^2} = -5 \frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

$$4 \frac{u_j^n - u_{j-1}^n}{5\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{10\Delta x} - \frac{\Delta x}{5} \frac{\partial^2 u}{\partial x^2} = -\frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

Sabe-se que $\frac{\partial^2 u}{\partial x^2} \cong \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{4\Delta x^2}$

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$

$$4 \frac{u_j^n - u_{j-1}^n}{5\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{10\Delta x} - \frac{\Delta x}{5} \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{4\Delta x^2} = -\frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$4 \frac{u_j^n - u_{j-1}^n}{5\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{10\Delta x} - \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20} = -\frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$4 \frac{u_j^n - u_{j-1}^n}{5\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{10\Delta x} - \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x^2} - \frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{4u_j^n - 4u_{j-1}^n}{5\Delta x} - \frac{(u_{j+2}^n - u_j^n)}{10\Delta x} - \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x} = -\frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{8u_j^n - 8u_{j-1}^n - u_{j+2}^n + u_j^n}{10\Delta x} - \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x} = -\frac{\partial u}{\partial x} - \frac{8\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$-\frac{8u_j^n - 8u_{j-1}^n - u_{j+2}^n + u_j^n}{10\Delta x} + \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x} = \frac{\partial u}{\partial x} - O[\Delta x^2]$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$-\frac{8u_j^n - 8u_{j-1}^n - u_{j+2}^n + u_j^n}{10\Delta x} + \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x} = \frac{\partial u}{\partial x} - O[\Delta x^2]$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2}^n - 9u_j^n + 8u_{j-1}^n}{10\Delta x} + \frac{u_{j+2}^n - 2u_j^n + u_{j-2}^n}{20\Delta x} + O[\Delta x^2]$$



Combinação das a Expansão de Taylor para 1 e 2 ponto de grade

terceira ordem

Avançado



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\frac{u_{j+1}^n - u_j^n}{\Delta x} = \frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

Multiplica por 4

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} = -4 \frac{\partial u}{\partial x} + \frac{4\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (DD)$$

soma DD da CC

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = -3 \frac{\partial u}{\partial x} + \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$4 \frac{u_j^n - u_{j-1}^n}{\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

$$4 \frac{u_j^n - u_{j-1}^n}{3\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{6\Delta x} - \frac{6\Delta x}{6} \frac{\partial^2 u}{\partial x^2} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (dd)$$

Sabe-se que $\frac{\partial^2 u}{\partial x^2} \cong \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2}$

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$

$$4 \frac{u_j^n - u_{j-1}^n}{3\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{6\Delta x} - \frac{\Delta x}{1} \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$4 \frac{u_j^n - u_{j-1}^n}{3\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{6\Delta x} - \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$4 \frac{u_j^n - u_{j-1}^n}{3\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{6\Delta x} - \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{4u_j^n - 4u_{j-1}^n}{3\Delta x} + \frac{(u_{j+2}^n - u_j^n)}{6\Delta x} - \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{8u_j^n - 8u_{j-1}^n + u_{j+2}^n - u_j^n}{6\Delta x} - \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{7u_j^n - 8u_{j-1}^n + u_{j+2}^n}{6\Delta x} - \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = -\frac{\partial u}{\partial x} + \frac{12\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4}$$

$$\frac{\partial u}{\partial x} = -\frac{9u_j^n - 8u_{j-1}^n - u_{j+2}^n}{6\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + 0[\Delta x^3]$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$\frac{\partial u}{\partial x} = -\frac{9u_j^n - 8u_{j-1}^n - u_{j+2}^n}{6\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + O[\Delta x^3]$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2}^n - 9u_j^n + 8u_{j-1}^n}{6\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + O[\Delta x^3]$$



Métodos de diferenças finitas.



Combinação das a Expansão de Taylor para 1 e 2 ponto de grade

3 opção terceira ordem

Avançado



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$\rightarrow u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$\rightarrow u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$



Dinâmica 23/09/2021 a 23/09/2021

Métodos de diferenças finitas.



Esquemas com diferenças centradas no espaço de terceira ordem:

$$\frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

Multiplica a BB por 4

$$4 \frac{u_{j-1} - u_j}{\Delta x} = -4 \frac{\partial u}{\partial x} + 4 \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} + O[(\Delta x)^4] \quad (AD)$$

soma a AD a eq. CC

$$4 \frac{u_{j-1} - u_j}{\Delta x} + \frac{u_{j+2} - u_j}{2\Delta x} = -3 \frac{\partial u}{\partial x} + \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{2\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

Sabe-se que $\frac{\partial^2 u}{\partial x^2} \cong \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2}$

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6\Delta x}{2} \left(\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} \right) = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \left(\frac{6u_{j+2} - 12u_j + 6u_{j-2}}{8\Delta x} \right) = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6u_{j+2} - 12u_j + 6u_{j-2}}{8\Delta x} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

$$\frac{4u_{j-1} - 4u_j}{3\Delta x} + \frac{u_{j+2} - u_j}{3\Delta x} - \frac{2u_{j+2} - 4u_j + 2u_{j-2}}{8\Delta x} = -\frac{\partial u}{\partial x} - O[\Delta x^3] \dots \quad (2D)$$

$$-\frac{4u_{j-1} - 4u_j}{3\Delta x} - \frac{u_{j+2} - u_j}{3\Delta x} + \frac{2u_{j+2} - 4u_j + 2u_{j-2}}{8\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots \quad (2D)$$

$$\frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots\dots \quad (2D)$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + \frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} - O[\Delta x^3] \dots\dots \quad (2D)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$u_{j+1}^n = u_j^n + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1B)$$

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$u_{j-2}^n = u_j^n - 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2C)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$u_{j-1}^n = u_j^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2B)$$

$$u_{j+2}^n = u_j^n + 2\Delta x \frac{\partial u}{\partial x} + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} + \frac{8\Delta x^3}{6} \frac{\partial^3 u}{\partial x^3} + \frac{16\Delta x^4}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (1C)$$

$$\rightarrow \frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\rightarrow \frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de terceira ordem:

$$\frac{u_{j-1}^n - u_j^n}{\Delta x} = -\frac{\partial u}{\partial x} + \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_{j+2}^n - u_j^n)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

Multiplica a BB por 4

$$4 \frac{u_{j-1} - u_j}{\Delta x} = -4 \frac{\partial u}{\partial x} + 4 \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} + O[(\Delta x)^4] \quad (AD)$$

soma a AD a eq. CC

$$4 \frac{u_{j-1} - u_j}{\Delta x} + \frac{u_{j+2} - u_j}{2\Delta x} = -3 \frac{\partial u}{\partial x} + \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{2\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6\Delta x}{2} \frac{\partial^2 u}{\partial x^2} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

Sabe-se que $\frac{\partial^2 u}{\partial x^2} \cong \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2}$

$$\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + \frac{4}{4!} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + O[(\Delta x)^4] \quad (6C)$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6\Delta x}{2} \left(\frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x^2} \right) = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \left(\frac{6u_{j+2} - 12u_j + 6u_{j-2}}{8\Delta x} \right) = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{4u_{j-1} - 4u_j}{\Delta x} + \frac{u_{j+2} - u_j}{\Delta x} - \frac{6u_{j+2} - 12u_j + 6u_{j-2}}{8\Delta x} = -3 \frac{\partial u}{\partial x} - \frac{4\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (2D)$$

$$\frac{4u_{j-1} - 4u_j}{3\Delta x} + \frac{u_{j+2} - u_j}{3\Delta x} - \frac{2u_{j+2} - 4u_j + 2u_{j-2}}{8\Delta x} = -\frac{\partial u}{\partial x} - O[\Delta x^3] \dots \quad (2D)$$

$$-\frac{4u_{j-1} - 4u_j}{3\Delta x} - \frac{u_{j+2} - u_j}{3\Delta x} + \frac{2u_{j+2} - 4u_j + 2u_{j-2}}{8\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots \quad (2D)$$

$$\frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots \quad (2D)$$



Métodos de diferenças finitas.

$$\frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + O[\Delta x^3] \dots\dots \quad (2D)$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + \frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} - O[\Delta x^3] \dots\dots \quad (2D)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de quarta ordem:



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de segunda ordem:

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (BB)$$

$$\frac{(u_j^n - u_{j-2}^n)}{2\Delta x} = \frac{\partial u}{\partial x} - \frac{2\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{4\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{8\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} \dots \quad (CC)$$

Multiplica a BB por 2

$$2 \frac{u_j - u_{j-1}}{\Delta x} = 2 \frac{\partial u}{\partial x} + 2 \frac{\Delta x}{2} \frac{\partial^2 u}{\partial x^2} + \frac{2\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{2\Delta x^3}{24} \frac{\partial^4 u}{\partial x^4} + O[(\Delta x)^4] \quad (AD)$$

Soma a AD a eq. CC

$$2 \frac{u_j - u_{j-1}}{\Delta x} + \frac{u_j - u_{j-2}}{2\Delta x} = 3 \frac{\partial u}{\partial x} + \frac{6\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} \dots$$

$$\frac{2}{3} \frac{u_j - u_{j-1}}{\Delta x} - \frac{1}{3} \frac{u_j - u_{j-2}}{2\Delta x} = \frac{\partial u}{\partial x} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$

$$\left(\frac{-u_j + 2u_j - 2u_{j-1} + u_{j-2}}{6\Delta x} \right) = \frac{\partial u}{\partial x} + \Delta x^2 \frac{\partial^3 u}{\partial x^3} \dots \quad (2D)$$



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de quarta ordem:

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{1}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O[(\Delta x)^4] \quad (4B)$$

$$\frac{u_{j+2} - u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} + \frac{4}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O[(\Delta x)^4] \quad (4C)$$

Multiplica a 4B por 4

$$4 \frac{u_{j+1} - u_{j-1}}{2\Delta x} = 4 \frac{\partial u}{\partial x} + \frac{4}{3!} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + O[(\Delta x)^4] \quad (1D)$$

Subtrai a 1D - 4C

$$4 \frac{u_{j+1} - u_{j-1}}{2\Delta x} - \frac{u_{j+2} - u_{j-2}}{4\Delta x} = 3 \frac{\partial u}{\partial x} + O[(\Delta x)^4]$$

$$\frac{4}{3} \frac{u_{j+1} - u_{j-1}}{2\Delta x} - \frac{1}{3} \frac{u_{j+2} - u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} \quad (2D)$$



Métodos de diferenças finitas.

Recurso Visual 1: Diagramas de Stencil — Pontos de Grade por Esquema

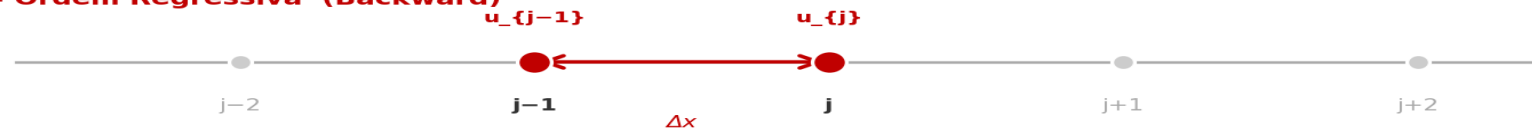
Diagramas de Stencil — Pontos de Grade Utilizados por Cada Esquema

1ª Ordem Progressiva (Forward)



$$du/dx \approx (u_{j+1} - u_j) / \Delta x$$

1ª Ordem Regressiva (Backward)



$$du/dx \approx (u_j - u_{j-1}) / \Delta x$$

2ª Ordem Centrada



$$du/dx \approx (u_{j+1} - u_{j-1}) / (2\Delta x)$$

4ª Ordem Centrada

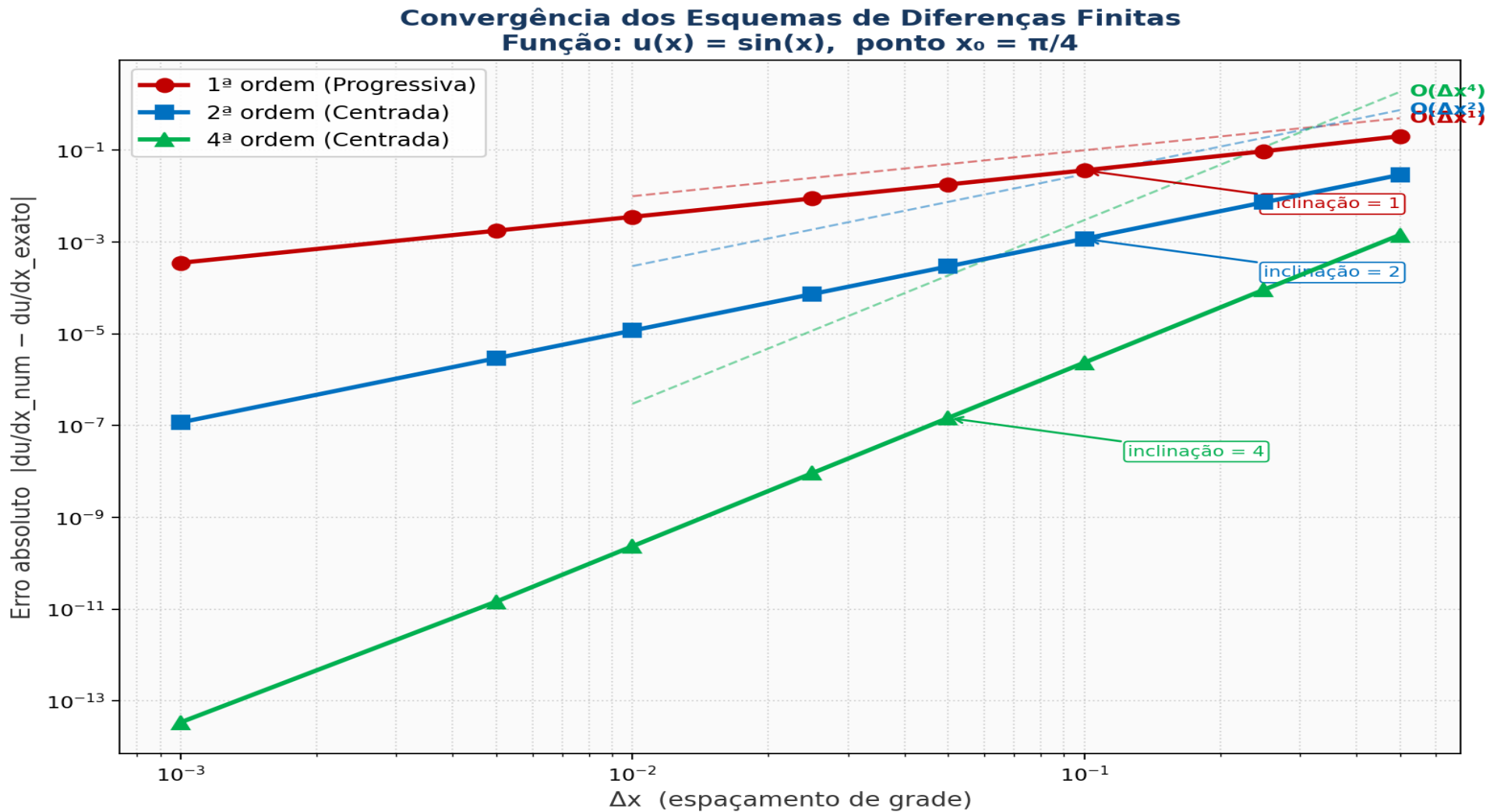


$$du/dx \approx (-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}) / (12\Delta x)$$



Métodos de diferenças finitas.

Recurso Visual 2: Gráfico de Convergência — Erro vs. Δx (escala log-log)





Conceito: Por que discretizar derivadas?

Motivação Física:

Em problemas de dinâmica de fluidos, transferência de calor e propagação de ondas, as equações envolvem derivadas contínuas que não têm solução analítica geral.

A discretização converte derivadas em operações algébricas entre valores em pontos de grade.

Exemplo: Equação de advecção-difusão unidimensional

$$\partial u / \partial t + c \cdot \partial u / \partial x = v \cdot \partial^2 u / \partial x^2$$

Onde c é a velocidade de advecção e v é o coeficiente de difusão.

Erro de truncamento:

Ao substituir $\partial u / \partial x$ pela diferença finita, cometemos um erro $O(\Delta x^n)$.

Quanto menor o Δx , menor o erro — mas maior o custo computacional.

O expoente n define a "ordem do método": 1ª ordem, 2ª ordem, 4ª ordem etc.



Métodos de diferenças finitas.



Exemplo Prático 1: Diferenças Progressiva e Regressiva (1ª Ordem)

Função de teste: $u(x) = \sin(x)$, $\partial u / \partial x = \cos(x)$, em $x_0 = \pi/4$

Diferença Progressiva (Forward): $du/dx \approx (u(x+\Delta x) - u(x)) / \Delta x + O(\Delta x)$

Com $\Delta x = 0.1$: $u(\pi/4+0.1) = \sin(0.885) \approx 0.7745$

$u(\pi/4) = \sin(0.785) \approx 0.7071$

Aproximação = $(0.7745 - 0.7071) / 0.1 = 0.674$

Valor exato = $\cos(\pi/4) \approx 0.7071$ → Erro $\approx 4.7\%$

Diferença Regressiva (Backward): $du/dx \approx (u(x) - u(x-\Delta x)) / \Delta x + O(\Delta x)$

Com $\Delta x = 0.1$: $u(\pi/4-0.1) = \sin(0.685) \approx 0.6330$

Aproximação = $(0.7071 - 0.6330) / 0.1 = 0.741$

Valor exato = $\cos(\pi/4) \approx 0.7071$ → Erro $\approx 4.8\%$

Conclusão: Ambas são 1ª ordem — o erro reduz linearmente com Δx .



Exemplo Prático 2: Diferença Centrada (2ª Ordem)

Diferença Centrada: $du/dx \approx (u(x+\Delta x) - u(x-\Delta x)) / (2 \cdot \Delta x) + O(\Delta x^2)$

Mesma função de teste: $u(x) = \sin(x)$, $x_0 = \pi/4$, $\Delta x = 0.1$

$$u(\pi/4 + 0.1) = \sin(0.885) \approx 0.7745$$

$$u(\pi/4 - 0.1) = \sin(0.685) \approx 0.6330$$

$$\text{Aproximação} = (0.7745 - 0.6330) / (2 \times 0.1) = 0.7075$$

$$\text{Valor exato} = \cos(\pi/4) \approx 0.7071 \rightarrow \text{Erro} \approx 0.06\%$$

Comparação direta com 1ª ordem (mesmo $\Delta x = 0.1$):

Progressiva: Erro $\approx 4.7\%$ | Regressiva: Erro $\approx 4.8\%$ | Centrada: Erro $\approx 0.06\%$

A diferença centrada é ~80x mais precisa com o mesmo espaçamento de grade!

Ao dobrar a resolução ($\Delta x \rightarrow \Delta x/2$): erro da centrada cai 4x ($O(\Delta x^2)$ vs $O(\Delta x)$ da 1ª ordem).



Estudo de Convergência: Reduzindo Δx

Função: $u(x) = \sin(x)$, $x_0 = \pi/4$, derivada exata = $\cos(\pi/4) = 0.70711$

Δx	Forward (1ª)	Centrada (2ª)	Centrada (4ª)
0.5000	0.0834 (11.8%)	0.0013 (0.18%)	$\sim 1.0e-5$
0.1000	0.0334 (4.7%)	0.0004 (0.06%)	$\sim 1.6e-7$
0.0500	0.0171 (2.4%)	$1.1e-4$ (0.015%)	$\sim 1.0e-8$
0.0100	0.0035 (0.5%)	$4.4e-6$ ($6.2e-4\%$)	$\sim 1.6e-11$

Razão de convergência observada:

Forward: dividir Δx por 2 $\rightarrow$ erro cai $\sim 2x$ (convergência linear, $O(\Delta x^1)$)

Centrada 2ª: dividir Δx por 2 $\rightarrow$ erro cai $\sim 4x$ ($O(\Delta x^2)$)

Centrada 4ª: dividir Δx por 2 $\rightarrow$ erro cai $\sim 16x$ ($O(\Delta x^4)$)



Exemplo Prático 3: Código Python — Comparando Esquemas

```
import numpy as np
import matplotlib.pyplot as plt

x0 = np.pi / 4      # Ponto de avaliação
exact = np.cos(x0)   # Derivada exata
dxs = [0.5, 0.1, 0.05, 0.01, 0.001] # Diferentes espaçamentos
for dx in dxs:
    u_f = np.sin(x0 + dx)      #  $u(x+\Delta x)$ 
    u_f2 = np.sin(x0 + 2*dx)    #  $u(x+\Delta x)$ 
    u_b = np.sin(x0 - dx)      #  $u(x-\Delta x)$ 
    u_b2 = np.sin(x0 - 2*dx)    #  $u(x-\Delta x)$ 
    u_0 = np.sin(x0)           #  $u(x)$ 
    fwd = (u_f - u_0) / dx      # 1ª ordem progressiva
    cent = (u_f - u_b) / (2*dx) # 2ª ordem centrada
    #cent4 = (-u_f*0+8*u_f-8*u_b) / (12*dx) # 4ª ordem centrada (simplif.)
    cent4 = (-u_f2 + 8*u_f - 8*u_b + u_b2) / (12*dx) # correto
    print(f"Δx={dx:.3f}: Fwd={abs(fwd-exact):.2e} Cent2={abs(cent-exact):.2e} Cent4={abs(cent4-exact):.2e}")
```



Métodos de diferenças finitas.

Resumo Comparativo dos Esquemas e Dicas Práticas

Esquema	Pontos usados	Ordem do Erro	Custo Comp.
Progressiva	u_j, u_{j+1}	$O(\Delta x^1)$	Baixo
Regressiva	u_j, u_{j-1}	$O(\Delta x^1)$	Baixo
Centrada 2ª ordem	u_{j-1}, u_{j+1}	$O(\Delta x^2)$	Médio
Centrada 3ª ordem	$u_{j-2} \dots u_{j+2}$	$O(\Delta x^3)$	Médio
Centrada 4ª ordem	$u_{j-2} \dots u_{j+2}$	$O(\Delta x^4)$	Alto

Dicas Práticas para Aplicações em MHD/Meteo/Oceano:

1. Use esquemas de 2ª ordem como padrão — boa relação precisão/custo.
2. Para gradientes intensos (frentes, turbulência), prefira 4ª ordem.
3. Esquemas de 1ª ordem introduzem difusão numérica — evite em long runs.
4. O código Fortran mostrado nos slides anteriores implementa esses esquemas.
5. Teste sempre a convergência: reduza Δx e verifique a taxa de decaimento do erro.



Métodos de diferenças finitas.

Esquemas com diferenças centradas no espaço de quarta ordem:

$$\frac{4}{3} \frac{u_{j+1} - u_{j-1}}{2\Delta x} - \frac{1}{3} \frac{u_{j+2} - u_{j-2}}{4\Delta x} = \frac{\partial u}{\partial x} \quad (2D)$$

$$\left(\frac{1}{3}\right) \left(\frac{4u_{j+1} - 4u_{j-1}}{2\Delta x} - \frac{u_{j+2} - u_{j-2}}{4\Delta x} \right) = \frac{\partial u}{\partial x} \quad (2D)$$

$$\left(\frac{1}{3}\right) \left(\frac{8u_{j+1} - 8u_{j-1} - u_{j+2} + u_{j-2}}{4\Delta x} \right) = \frac{\partial u}{\partial x} \quad (2D)$$

$$\left(\frac{-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}}{12\Delta x} \right) = \frac{\partial u}{\partial x} \quad (2D)$$



Resumo

$$\frac{\partial u}{\partial x} = \frac{u_{j+1} - u_{j-1}}{2\Delta x} - O[\Delta x^2]$$

$$\frac{\partial u}{\partial x} = \left(\frac{-2u_{j-1} + u_j + u_{j+2}}{6\Delta x} \right) - O[\Delta x^2]$$

$$\frac{\partial u}{\partial x} = \left(\frac{4u_{j-1} - 3u_j - u_{j+2}}{6\Delta x} \right) - O[\Delta x^2]$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2}^n - 9u_j^n + 8u_{j-1}^n}{10\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{20\Delta x} + O[\Delta x^2]$$

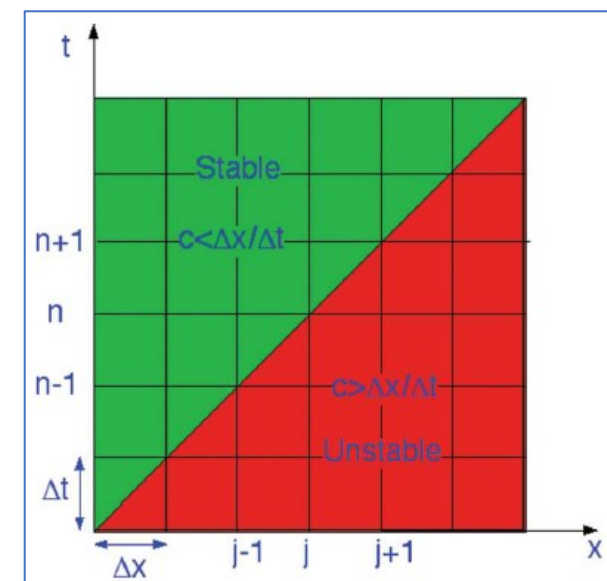
$$\frac{\partial u}{\partial x} = \frac{u_{j+2}^n - 9u_j^n + 8u_{j-1}^n}{6\Delta x} + \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + O[\Delta x^3]$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+2} - 2u_j + u_{j-2}}{4\Delta x} + \frac{-4u_{j-1} + 4u_j - u_{j+2} + u_j}{3\Delta x} - O[\Delta x^3] \dots (2D)$$

$$\frac{\partial u}{\partial x} = \left(\frac{-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}}{12\Delta x} \right) + O[\Delta x^4] \dots (2D)$$

$$\frac{\partial u}{\partial x} = \frac{u_j^n - u_{j-1}^n}{\Delta x} + O[\Delta x^1]$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+1}^n - u_j^n}{\Delta x} + O[\Delta x^1]$$





Resumo

Module ModClassDerivada

IMPLICIT NONE

INTEGER :: xdim

REAL, ALLOCATABLE :: f(:)

REAL, ALLOCATABLE :: ff(:)

REAL, ALLOCATABLE :: x(:)

REAL, ALLOCATABLE :: df2dx(:)

REAL :: dx

REAL :: Length

REAL, PARAMETER :: Pi=3.14

PUBLIC :: InitClass,runDeriv,finalizeClassDerivada

CONTAINS

SUBROUTINE InitClass

INTEGER :: i,xb,xc,xf

Length=2*Pi

xdim =100

Dx= Length/REAL(xdim)

allocate(ff(0:xdim)); ff(0:xdim) =0.0

allocate(df2dx(0:xdim)); df2dx(0:xdim) =0.0

allocate(f(0:xdim)); f(0:xdim) =0.0

allocate(x(0:xdim)); x(0:xdim) =0.0

DO i=1,xdim

CALL index(i,xdim,xb,xc,xf)

x(xc) = x(xb) + Dx

END DO

DO i=0,xdim

CALL index(i,xdim,xb,xc,xf)

f (xc) = sin(x(xc))

ff(xc) = cos(x(xc))

END DO

END SUBROUTINE InitClass

SUBROUTINE runDeriv()

INTEGER :: i,xb,xc,xf

DO i=0,xdim

CALL index(i,xdim,xb,xc,xf)

! df2dx(i) = ((F(xf) - F(xb))/(2.0*Dx))

df2dx(i) = ((F(xf) - F(xc))/(Dx))

!df2dx(i) = ((F(xb) - F(xc))/(Dx))

END DO

PRINT*, 'Effective Error=',SUM(df2dx(1:xdim-1)-ff(1:xdim-1))/SUM(ff(1:xdim-1))

CALL WriteBinary()

CALL WriteCTL()

END SUBROUTINE runDeriv

SUBROUTINE WriteBinary()

INTEGER :: lrec

INQUIRE(IOLENGTH=lrec)df2dx(1:xdim-1)

OPEN(1,FILE='runDeriv.bin',ACCESS='DIRECT',FORM='UNFORMATTED', RECL=lrec,&
STATUS='UNKNOWN',ACTION='WRITE')

WRITE(1,rec=1)df2dx(1:xdim-1)

WRITE(1,rec=2)ff (1:xdim-1)

WRITE(1,rec=3)f (1:xdim-1)

CLOSE(1,STATUS='KEEP')

END SUBROUTINE WriteBinary



Resumo

```
SUBROUTINE WriteCTL()  
  INTEGER :: i  
  OPEN(1,FILE='runDeriv.ctl',ACCESS='SEQUENTIAL',FORM='FORMATTED', &  
    STATUS='UNKNOWN',ACTION='WRITE')  
  
  WRITE(1,'(A6,A12    )')'dset ^','runDeriv.bin'  
  WRITE(1,'(A        )')'*options big_endian'  
  WRITE(1,'(A6,A10    )')'undef ','-999.99999'  
  WRITE(1,'(A6,I5,A8    )')'xdef ',xdim-1,' levels '  
  WRITE(1,'(10F15.5)')(x(i),i=1,xdim-1)  
  WRITE(1,'(A6,A3,A8,A7 ,A4)')'ydef ', ' 1 ',' linear ', ' -90.0 ', ' 1.0'  
  WRITE(1,'(A6,A3,A8,A14,A4)')'tdef ', ' 1 ',' linear ', ' 00z01apr2014 ', ' 1hr'  
  WRITE(1,'(A6,A3,A8,A4  )')'zdef ', ' 1 ',' levels ', '1000'  
  WRITE(1,'(A6,A3      )')'VARS ', ' 3 '  
  WRITE(1,'(A6,A23      )')'Fnum ', '0 99 derivada numerica '  
  WRITE(1,'(A6,A23      )')'Fana ', '0 99 derivada analitica '  
  WRITE(1,'(A6,A23      )')'Fori ', '0 99 funcao original '  
  WRITE(1,'(A7          )')'ENDVARS'  
END SUBROUTINE WriteCTL
```

```
SUBROUTINE index(i,ldim,xb,xc,xf)  
  IMPLICIT NONE  
  INTEGER, INTENT(IN      ) :: i  
  INTEGER, INTENT(IN      ) :: ldim  
  INTEGER, INTENT(OUT ) :: xb,xc,xf  
  IF(i==0) THEN  
    xb=ldim  
    xc=i  
    xf=i+1  
  ELSE IF(i==ldim)THEN  
    xb=ldim-1  
    xc=ldim  
    xf=0  
  ELSE  
    xb=i-1  
    xc=i  
    xf=i+1  
  END IF  
END SUBROUTINE index  
  
SUBROUTINE finalizeClassDerivada()  
  DEALLOCATE(F)  
  DEALLOCATE(FF)  
  DEALLOCATE(df2dx)  
  DEALLOCATE(x)  
  
END SUBROUTINE finalizeClassDerivada  
  
END Module ModClassDerivada  
!  
=====
```



Resumo



```
!=====
PROGRAM Main
  USE ModClassDerivada, Only: InitClass,runDeriv,&
    finalizeClassDerivada
  IMPLICIT NONE

  CALL Init()
  CALL run()
  CALL finalize()

CONTAINS

SUBROUTINE Init()
  CALL InitClass()
END SUBROUTINE Init

SUBROUTINE run()
  CALL runDeriv()
END SUBROUTINE run

SUBROUTINE finalize()
  CALL finalizeClassDerivada()
END SUBROUTINE finalize

END PROGRAM Main
!=====
```